Study Group Formation

Unsupervised Clustering Project Report

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Project overview

This project uses student study-related data to identify natural profiles and form study groups. K-Means is the primary clustering method and Ward Hierarchical Clustering is the comparison method. The Python and Streamlit implementations support two grouping modes: default K-Means grouping when no matching attributes are selected, and hard attribute matching when the user chooses target attributes and values.

The output is a recommendation for an instructor. It should be reviewed alongside classroom context, because the dataset is synthetic and the model does not measure interpersonal fit or actual collaboration outcomes.

Part 1 Dataset and data dictionary

The project dataset contains 60 synthetic student records. Each record includes a pseudonymous identifier, display information, study interests, four skill measures, weekly availability, and preferred group size.

Field	Role	Description
student_id	Reference/display	Unique pseudonymous identifier; excluded from modeling.
display_name	Reference/display	Human-readable label; excluded from numeric distance calculations.
interest_tags	Reference/display	Interest tags retained for interpretation and display.
programming, design, communication, analytics	Analytical input	Self-rated skill scores from 1 to 10.
availability_hours	Analytical input	Weekly availability from 2 to 25 hours.
preferred_group_size	Analytical input	Preferred group size from 2 to 8.
kmeans_cluster	Generated output	K-Means assignment for the analytical comparison run.

hierarchical_cluster	Generated output	Ward hierarchical assignment for the comparison run.
formed_group	Generated output	Exact-size group number produced by the interface grouping logic.

Missing values are checked in Python and imputed with numeric medians for batch analysis. Interface inputs are constrained to valid ranges and incomplete or invalid cases produce a validation message.

Part 2 Data preparation

- Check that student_id values are unique and retain identifiers only for display.
- Keep six numerical model features: programming, design, communication, analytics, availability_hours, and preferred_group_size.
- Check missing values and duplicates before modeling.
- Impute numeric missing values with the median when needed.
- Standardize the numerical features before distance-based clustering.

Part 3 KNIME workflow

The corrected KNIME workflow uses the following main branch:

CSV Reader -> Column Filter -> Missing Value -> Normalizer -> k-Means -> Scatter Plot

The corrected comparison branch is:

Normalizer -> Hierarchical Clustering

The Column Filter now excludes student_id, display_name, and interest_tags from the model input. The Normalizer scales the six numerical fields. K-Means is configured with k=3 and seed 42. Hierarchical Clustering uses the same six normalized fields, Euclidean distance, Ward linkage, and three clusters. The scatter plot uses programming on the x-axis and communication on the y-axis.

For exact-size matching in the interface mode, the matching rows are filtered first, then complete groups are formed only from the matching rows. If there are not enough matching records, unrelated students are not added.

Part 4 Python implementation

The Python implementation uses pandas for loading and validation, StandardScaler for feature scaling, KMeans for the primary model, AgglomerativeClustering for comparison, and silhouette_score for evaluation.

The batch comparison run reports the following observed silhouette scores:

Method	Silhouette score	Interpretation
K-Means	0.476	Primary method; stronger separation in the observed run.
Hierarchical Ward	0.460	Comparison method; slightly lower separation in the observed run.

The complete Python implementation is included in study_group_colab.py and the Colab-ready notebook is included in study_group_clustering.ipynb.

Part 5 Streamlit interface

The interface uses a blue and light-blue palette and collects skill values, availability, and the number of students per formed group. It also includes a Students Master List button that displays the complete dataset.

Default mode: leave Attributes to match empty. The app performs the normal K-Means grouping across the full dataset.

Selected-attribute mode: choose one or more attributes, such as Programming skill and Communication skill. The app keeps only students within the configured tolerance of the selected values. Those students alone can enter the formed groups. If there are too few matches for a complete group, the app shows the matching students and a warning instead of adding unrelated students.

Part 6 Testing and reflection

Test case	Expected result
Normal	Default K-Means grouping runs when no attributes are selected.
Selected attributes	Only students matching every selected attribute can enter groups.
Missing value	Batch Python imputes numeric medians; the interface validates incomplete input.
Boundary	Skills remain 1 to 10, availability 2 to 25, and group size 2 to 8.
Uncertain	Values near the tolerance boundary may change membership.
Insufficient matches	No unrelated students are added; matching rows and a warning are shown.

Critical-thinking questions

1. The number of clusters should be selected using silhouette score, cluster-size balance, domain usefulness, and whether the groups are actionable. A high score alone is not enough.

2. Groups are meaningfully different when standardized skill profiles, availability, or preferred group size change likely roles, meeting feasibility, or peer-support needs.

3. If assignments change after an update, rerun preprocessing and both algorithms, record the changed assignments and scores, and investigate data drift, missing values, outliers, and model instability.

Interpretation and limitations

The clustering results provide evidence of natural student profiles rather than definitive labels. K-Means produced slightly stronger separation than Ward Hierarchical Clustering in the observed comparison. The interface allows instructors to use either a default data-driven grouping or targeted attribute matching.

Limitations include synthetic and self-reported data, a cross-sectional design, sensitivity of K-Means to scaling, initialization, outliers, and k, dependence of hierarchical results on linkage and distance, and the fact that the model does not measure personality, learning preferences, interpersonal fit, or collaboration outcomes. Instructor review remains necessary.

Submission files

- study_group_students.csv
- data_dictionary.md
- KNIME_MIDTERM_REQUIREMENT.knwf
- knime_workflow.md
- knime_workflow_blueprint.json
- study_group_colab.py
- study_group_clustering.ipynb
- streamlit_app.py
- comparison_table.csv
- comparison_chart.svg
- Study_Group_Formation_Report.pdf